Determination of the Absolute Zero Point

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The goal of this experiment is to approximate the absolute zero temperature. Since there is no absolute negative pressure and Volume the formular $pV = Nk_BT$ implies that there is an absolute zero temperature. To measure approximate we made an experiment which can measure The helium pressure at different temperature. With these values we could approximate the absolute zero temperature to $(-273 \pm 5)^\circ$C. Which lines within the literature value -273.15° Celsius [1]. Using this result we were now able to approximate the temperature of N_2.

1 Introduction

The absolute zero point is a fundemental fixed point which has the lowest possible temperature t_0. This temperature t_0 is achived when all the particles of the system have 0 velocity. The Kelvin temperature scale is named after the physicist Lord Kelvin (William Thomson), who proposed an absolute temperature scale in 1848. The scale starts at absolute zero (0 K), the lowest possible thermodynamic temperature. Temperatures on the Kelvin scale are therefore non-negative

The Temperature is also contained in the ideal-gas-law. Here N is the number of particles, k_B is the Boltzmann constant, T is Temperature, p is pressure, and V is the Volume.

$$pV = Nk_BT \tag{1.1}$$

Helium (He) behaves relatively close to the ideal gas law, therefore this Equation (1.1) can be used as a good approximation. So our calculations are mainly based on this relation.

2 Experiment

The experimental (see Figure 1) allows us to measure differents voltages which are linearly related to the pressure in the glas bulb. In Section 3.1 we see more precisely how to calculate the pressure given the voltage. This setup is connected to a pump which allows us by opening valve 2 to create a pressure close to a vacuum. Thourgh valve 3 and 7 we were able to fill the glas bulb up with Helium.
Firtsly we wann to calibrate the voltmeter. For this we measure two situations in which we know the pressure and temperature. We looked at the voltage with room temperature and room pressure and as a secon measurement

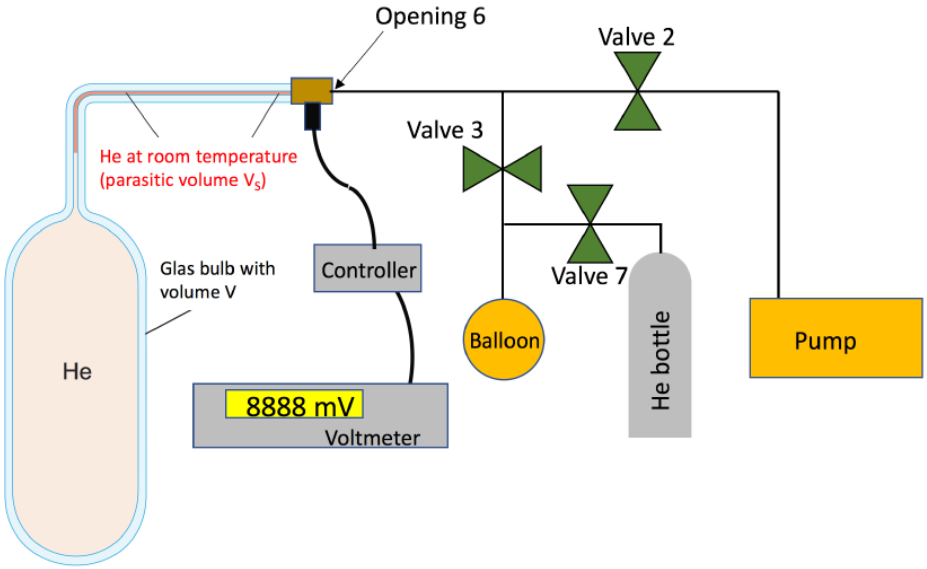


Figure 1: Shows the experimental setup. [2]

3 Results

3.1 Calibration

For the calibration we measured the laboratory temperature t_l. We also measured the reading from the barometer $\tilde{p}_l$. Aswell as the output-signal of the pressure sensor when exposed to laboratory pressure U_l. Then we measured the output-signal at low pressure U_2. This gave:

$$\begin{aligned} t_l &= (22 \pm 1)°C \\ \tilde{p}_l &= (716 \pm 1)\ \text{mmHg} \\ U_l &= (128.0 \pm 0.2)\ \text{mV} \\ U_2 &= (-5.3 \pm 0.2)\ \text{mV} \end{aligned} \tag{3.1}$$

Far all temperatures we estimated $\sigma_t = 1°C$. Since the smallest scale on the Barometer is ± 1 mmHg, we got $\sigma_p = 1$ mmHg. For all output voltages we used $\sigma_U = 0.2$ mV. The σ_U comes from the manual [2] where there is the 0.05% full scale $= 0.075$ mV percicion error and another error due to temperature difference of $0.08 \frac{\text{mV}}{°\text{C}}$. Since we estimated $\sigma_t = 1°C$ we got $\sigma_U = 0.075\ \text{mV} + 0.08\ \text{mV} \approx 0.2\ \text{mV}$.

3.2 Absolute Zero

The measurements to determine the absolute zero temperature are the output signals of the pressure sensor. We use U_k for boiling water and U_e for freezing water. The results are:

$$\begin{aligned} U_k &= (127.6 \pm 0.2)\ \text{mV} \\ U_e &= (92.5 \pm 0.2)\ \text{mV} \end{aligned} \tag{3.2}$$

3.3 Liquid Nitrogen

The measurment for liquid Nitrogen (ln2) is U_{ln2}. This is the output signal from the pressure sensor when submerged in boiling liquid Nitrogen.

$$U_{\text{ln2}} = (24.7 \pm 0.2)\ \text{mV} \tag{3.3}$$

4 Data Analysis

4.1 Calibration

From the barometer reading $\tilde{p}_l$ and laboratory temperature t_l we calculate the pressure in the laboratory p_l

$$p_l = (\tilde{p}_l - \text{correction}(t_l)) * \frac{133.322}{100} = ((716 \pm 1) - 3) * \frac{133.322}{100}\ \text{mbar} = (951 \pm 1)\ \text{mbar} \tag{4.1}$$

We used the value of U_k for U_l. So $U_l = (127.6 \pm 0.2)\ \text{mV}$ for all calculations (see Section 5). The value for the low pressure is per assumption [2]:

$$p_2 = (0.1 \pm 0.1)\ \text{mbar} \tag{4.2}$$

From here we fitted a linear function of the form:

$$p(U) = p_0 + CU \tag{4.3}$$

through (p_l, U_l) and (p_2, U_2). We calculated:

$$\begin{aligned} C &= \frac{p_l - p_2}{U_l - U_2} = (7.16 \pm 0.02)\frac{\text{bar}}{V} \\ p_0 &= p_l - \frac{p_l - p_2}{1 - \frac{U_2}{U_l}} = (0.038 \pm 0.001)\ \text{bar} \end{aligned} \tag{4.4}$$

For the error propagations we always used a function of the starting parameters to calculate the values. We used the scipy Jacobian [3] function to calculate all the derivatives. So for C it is:

$$
\begin{aligned}
z_C &= (p_l, p_2, U_l, U_2) \\
\sigma_{z_C} &= \left(\sigma_{p_l}, \sigma_{p_2}, \sigma_{U_l}, \sigma_{U_2}\right) = (0.001\ \text{bar}, 0.0001\ \text{bar}, 0.2\ \text{mV}, 0.2\ \text{mV}) \\
f_C(z_C) &:= \frac{p_l - p_2}{U_l - U_2} \\
\sigma_C &= \sqrt{\left(J_{f_C}(z_C)^2\right)_{1\times 4}\left(\sigma^2_{z_C}\right)^T_{4\times 1}} = 0.02\frac{\text{bar}}{V}
\end{aligned}
\tag{4.5}
$$

Where $J_{f_C}(z_C)^2 = \left(\left(\partial_{p_l} f_C(z_C)\right)^2, \left(\partial_{p_2} f_C(z_C)\right)^2, \left(\partial_{U_l} f_C(z_C)\right)^2, \left(\partial_{U_2} f_C(z_C)\right)^2\right)$ and $\sigma^2_{z_C} = \left(\sigma^2_{p_l}, \sigma^2_{p_2}, \sigma^2_{U_l}, \sigma^2_{U_2}\right)$. We used the same prcedure for all propagated errors. The exact calculations can be found in the python program at Section A.1. We used AI-tools to help debug the python program. We always made the function depend only on the measured values and the values assumed and not on previously caluclated values. We did this to minimize the effect of correlation. Since previously calculated values can depend on the same starting parameters they correlate. To account for correlation the error propagation gets more complicated and one has to estimate the correlation between the values. This is why we always directly calculated from the original values, so we can assume that they are uncorrelated.

So for σ_{p_0} we have $\sigma_{p_0} = \sqrt{\left(J_{f_{p_0}}\left(z_{p_0}\right)^2\right)_{1\times 4}\left(\sigma^2_{z_{p_0}}\right)^T_{4\times 1}}$ with $z_{p_0} = z_{p_C} \Rightarrow \sigma_{z_{p_0}} = \sigma_{z_C}$ and $f_{p_0}\left(z_{p_0}\right) = p_l - \frac{p_l - p_2}{1 - \frac{U_2}{U_l}}$

The calibration gives the following calibration-curve:

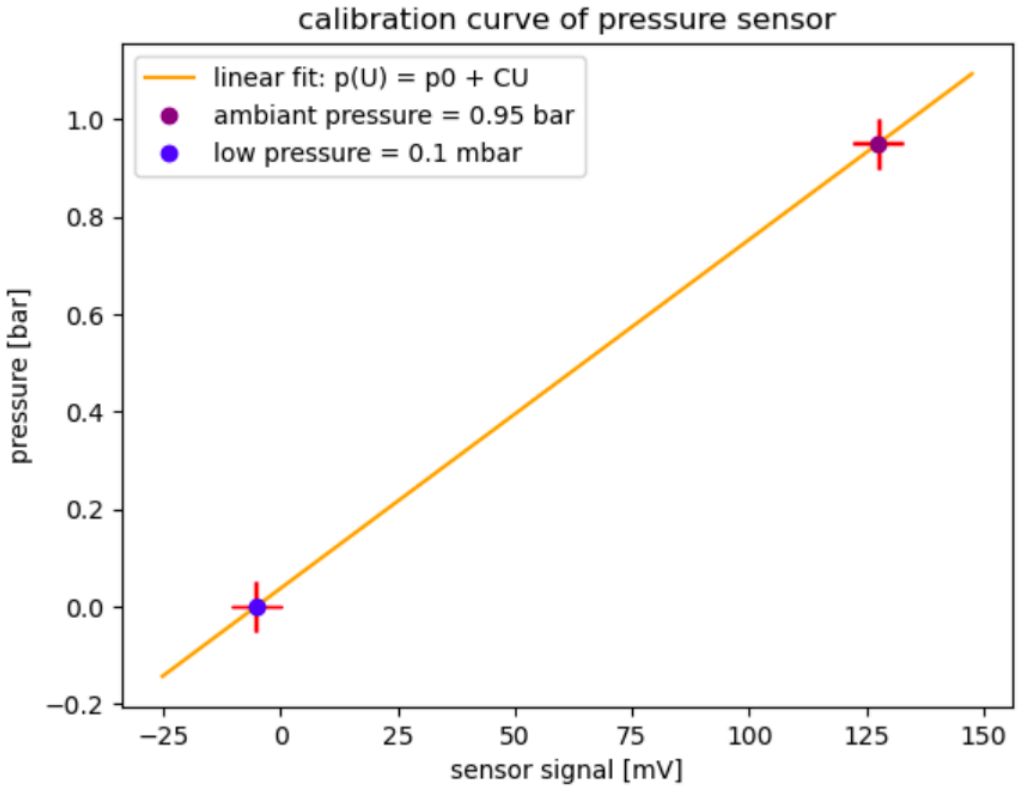


Figure 2: The two calibration points. (p_l, U_l) is purple and (p_2, U_2) is blue. Orange line is $p(U) = p_0 + CU$. Note: The red lines are not the error bars. They are the caps of the error bars. The error bars are smaller than the points and thus barely visible.

4.2 Absolute Zero

From the calibration curve and U_k and U_e we obtain the pressure p_k and p_e at boiling temperature (water) t_k and at freezing temperature t_e.

$$
\begin{aligned}
p_k &= p(U_k) = (951 \pm 2)\ \text{mbar} \\
p_e &= p(U_e) = (700 \pm 2)\ \text{mbar} \\
t_k &= (98 \pm 1)°C \\
t_e &= (0 \pm 1)°C
\end{aligned}
\tag{4.6}
$$

We obtained t_k through a table in the laboratory where the boiling temperature of water is denoted in dependance of p_l. For the freezing temperature t_e we have $t_e = 0$ per assumption. We can now also plot the pressure/temperature dependance:

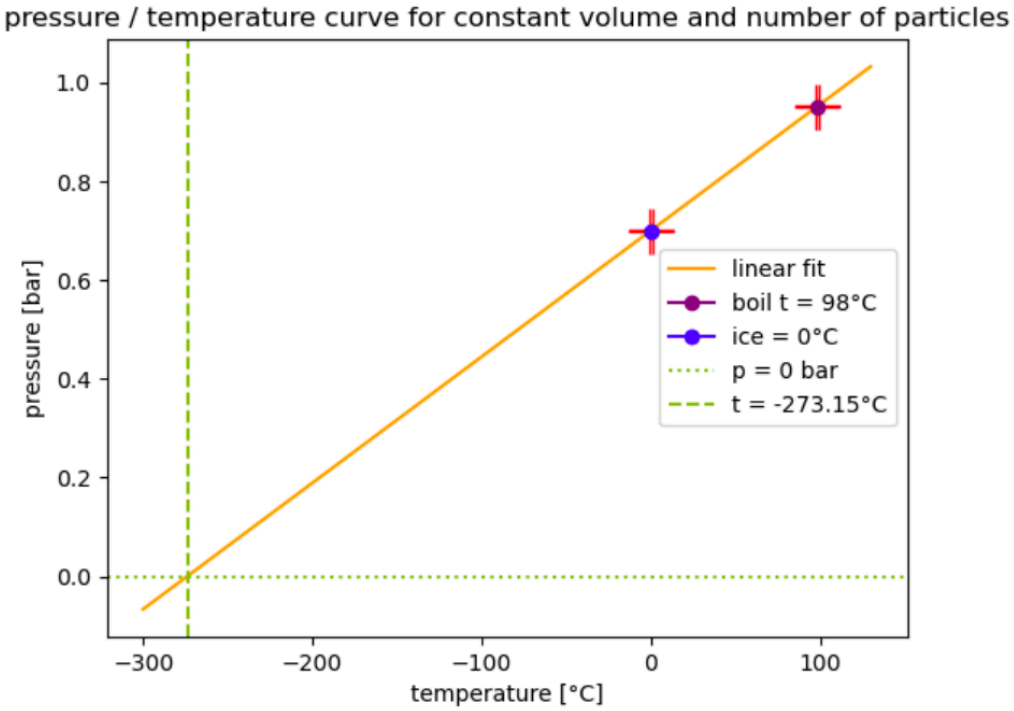


Figure 3: The pruple point is the measured point of (t_k, p_k). The blue point is the measured point of (t_e, p_e). The orange line is a linear fit. From extrapolating the orange curve, one gets t_0'. The green dashed line at $t = -273.15°C$ represents the literature value of absolute zero temperature. It does not represent the experimental value of t_0'. Note: The red lines are not the error bars. They are the caps of the error bars. The error bars are smaller than the points and thus barely visible.

From the obtained values we can caluclated the temperature of absolute zero. t_0' is a simpler approximative calculation. t_0 is more accurate since it takes the parasitic volume V_S into account (see Figure 1).

$$t_0' = -\frac{p_e}{p_k - p_e} t_k = (-274 \pm 3)°C \qquad [2] \tag{4.7}$$

we define:

$$\begin{gathered} \varepsilon = 10^{-3} \\ \gamma = 10^{-5}\ /°\mathrm{C} \\ a = (1+\varepsilon)p_e - (1+\varepsilon+\gamma t_k)p_k \\ b = \varepsilon(p_k - p_e)t_k + (1+\gamma t_k)p_k t_l - p_e(t_l + t_k) \\ c = p_e t_l t_k \qquad [2] \end{gathered} \tag{4.8}$$

this gives:

$$t_0 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} = (-273 \pm 5)°C \qquad [2] \tag{4.9}$$

4.3 Liquid Nitrogen

From the calibration curve and the measurment U_{ln2} we obtain the value for the pressure p_{ln2} at temperature of liquid nitrogen (t_{ln2})

$$p_{\mathrm{ln2}} = (214 \pm 1)\ \mathrm{mBar} \tag{4.10}$$

Again there is a simpler approximation for the temperature of liquid Nitrogen (t'_{ln2}). Aswell as a more complicated and more accurate value t_{ln2}. We have:

$$t'_{\ln 2} = \left(1 - \frac{p_{\ln 2}}{p_e}\right) t_0 = (-189 \pm 4)°C \qquad [2] \tag{4.11}$$

we define:

$$\begin{gathered} \varepsilon = 10^{-3} \\ \gamma = 10^{-5}\ /°\mathrm{C} \\ A = \frac{p_e}{t_e - t_0} + \varepsilon \frac{p_e - p_{\ln 2}}{t_l - t_0} \qquad [2] \end{gathered} \tag{4.12}$$

this gives:

$$t_{\ln 2} = \frac{A t_0 + p_{\ln 2}}{A - \gamma p_{\ln 2}} = (-189 \pm 4)°C \qquad [2] \tag{4.13}$$

5 Discussion

Whilst measuring the value for U_l we did not disconnect the pressurized hose from the system. We instead opened the valves to expose the sensor to the remaining pressure in the hose system. This pressure is not the same as the ambiant laboratory pressure. Whilst heating up the helium in the steam bath, the hose was fully disconnected and the valve at position 6 (see in Figure 1) was open. This means that the sensor was exposed to laboratory pressure. We also noticed that the reading of the sensor was stable in this configuration. We therefore concluded that the measurment of U_k better represents the laboratory pressure U_l. We therefore replaced the measured value of $U_l = (128.0 \pm 0.2)\ \mathrm{mV}$ with $U_k = (127.6 \pm 0.2)\ \mathrm{mV}$. So in all calculations: $U_l = U_k = (127.6 \pm 0.2)\ \mathrm{mV}$.

The literature value of the absolute zero temperature ($-273.15°C$) lies within the experimental result range of $t_0 = (-273 \pm 5)°C$.

The literature value of the temperature of liquid Nitrogen is: $-195.8°C$. This value lies outside the experimentally determined range of $t_{\ln 2} = (-189 \pm 4)°C$. So there is likely a systematic error. One possible source of error is leakage of gas. There could be small amounts of gas entering the lower pressure system, which would increase the measured pressure. This would lead to a higher experimentally determined boiling temperature, like in our case. The measurment of the liquid Nitrogen was taken last, so in this case it would not be contradictury to an accurate measurment of absolute zero temperature t_0. Another source could be that the temperature of the helium never fully reached the temperature of liquid Nitrogen. In this case it would be warmer, since we never cooled it more down, than with liquid Nitrogen. This again would lead to a too high experimental value of $t_{\ln 2}$. Also we noticed a drop in output voltage of the pressure sensor, after submerging the glas bulb (Figure 1) in liquid Nitrogen (this is the expected behaviour). After the voltage dropped, it immediatly began to rise again. So the output voltage never stabilized, which made an accurate measurment difficult. We thus think, that an inaccurate measurment is the most likely cause of error.

6 Conclusion

conclusion

A Appendix

A.1 python skript

Bibliography

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[3] SciPy, “jacobian.” Accessed: Nov. 06, 2026. [Online]. Available: https://docs.scipy.org/doc/scipy/reference/generated/scipy.differentiate.jacobian.html